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Magnetic Skyrmions

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$$H = H_J + H_D$$

Heisenberg exchange Hamiltonian : $H_J = -J \sum_r \vec{S}_r \cdot (\vec{S}_{r+\hat{x}} + \vec{S}_{r+\hat{y}}) - H_Z \sum_r \vec{S}_r$

Dzyaloshinskii-Moriya interaction Hamiltonian : $H_D = D \sum_r \vec{S}_r \times \vec{S}_{r+\hat{x}} \cdot \hat{x} + \vec{S}_r \times \vec{S}_{r+\hat{y}} \cdot \hat{y}$